30-Minute Student Task

Build Your Own Python Library Management System

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Programming Language	Python

ASSESSMENT

A simple menu-driven Library Management System was developed in Python. The application allows the user to add books, view available books, search for a book, issue a book, and return a book. The program uses basic Python concepts and provides an easy interface for managing library records.

INTRODUCTION

The Library Management System is a Python-based application developed to manage book records in a simple and efficient way. The system stores book information and allows the user to perform common library operations through a menu-driven interface.

This project demonstrates important Python programming concepts such as dictionaries, lists, loops, conditional statements, functions, input handling, and basic record management.

OBJECTIVE

The main objectives of the Library Management System are:

• To maintain library book records efficiently.

• To add new books to the library.

• To display all stored books.

• To search for a particular book.

• To issue a book to a student.

• To return an issued book.

• To demonstrate practical implementation of Python programming concepts.

PYTHON SOURCE CODE

```
books = {
    101: {"title": "Python Basics", "author": "John
Smith", "status": "Available"},
    102: {"title": "Data Structures", "author": "Mark
Lee", "status": "Available"}
}

def add_book():
    book_id = int(input("Enter Book ID: "))
    title = input("Enter Book Title: ")
    author = input("Enter Author: ")

    books[book_id] = {
        "title": title,
        "author": author,
        "status": "Available"
    }

    print("Book added successfully.")

def view_books():
    print("\n LIBRARY BOOKS")
    for book_id, book in books.items():
        print(f"{book_id} | {book['title']} |
{book['author']} | {book['status']}")

def search_book():
    book_id = int(input("Enter Book ID to search: "))

    if book_id in books:
        book = books[book_id]
        print("\nBook Found")
        print("Title :", book["title"])
        print("Author:", book["author"])
        print("Status:", book["status"])
    else:
        print("Book not found.")
```

```
def issue_book():
    book_id = int(input("Enter Book ID to issue: "))

    if book_id in books and books[book_id]["status"] ==
"Available":
        books[book_id]["status"] = "Issued"
        print("Book issued successfully.")
    else:
        print("Book not available.")

def return_book():
    book_id = int(input("Enter Book ID to return: "))

    if book_id in books:
        books[book_id]["status"] = "Available"
        print("Book returned successfully.")
    else:
        print("Book not found.")

while True:
    print("\n===LIBRARY MANAGEMENT SYSTEM
==========")
    print("1. Add Book")
    print("2. View Books")
    print("3. Search Book")
    print("4. Issue Book")
    print("5. Return Book")
    print("6. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        add_book()

    elif choice == "2":
        view_books()

    elif choice == "3":
        search_book()

    elif choice == "4":
```

```
        issue_book()

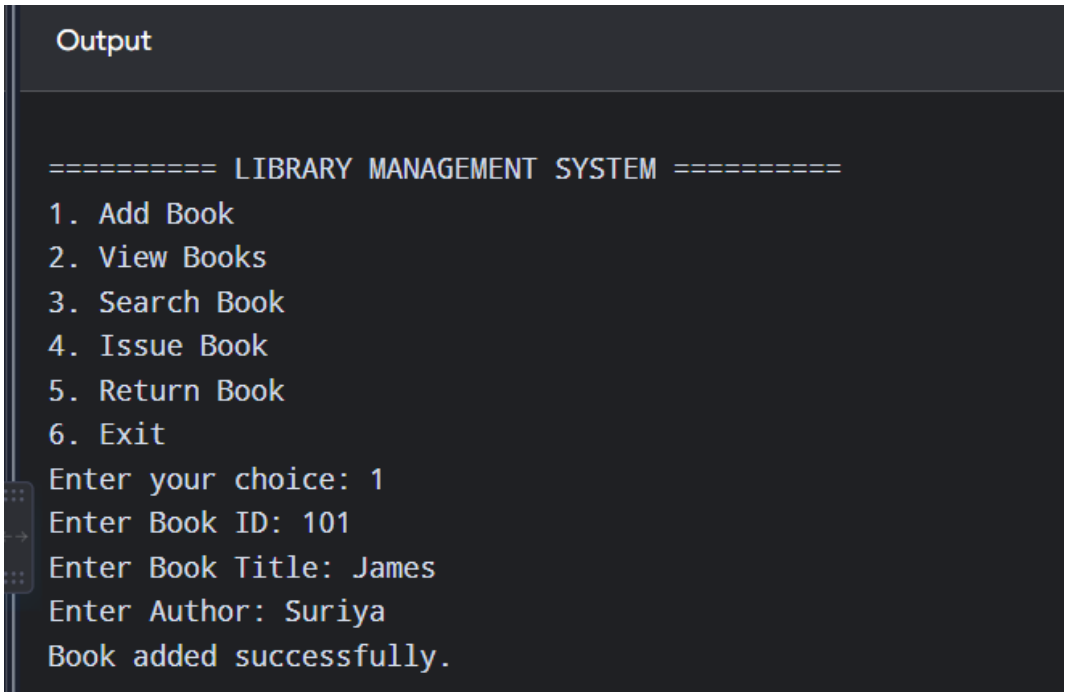
    elif choice == "5":
        return_book()

    elif choice == "6":
        print("Thank You!")
        break

    else:
        print("Invalid Choice.")
```

Execution and Test Output

1) Add Book



2) View Books

```
Output

 LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 2

===== LIBRARY BOOKS =====
101 | Python Basics | John Smith | Available
102 | Data Structures | Mark Lee | Available
```

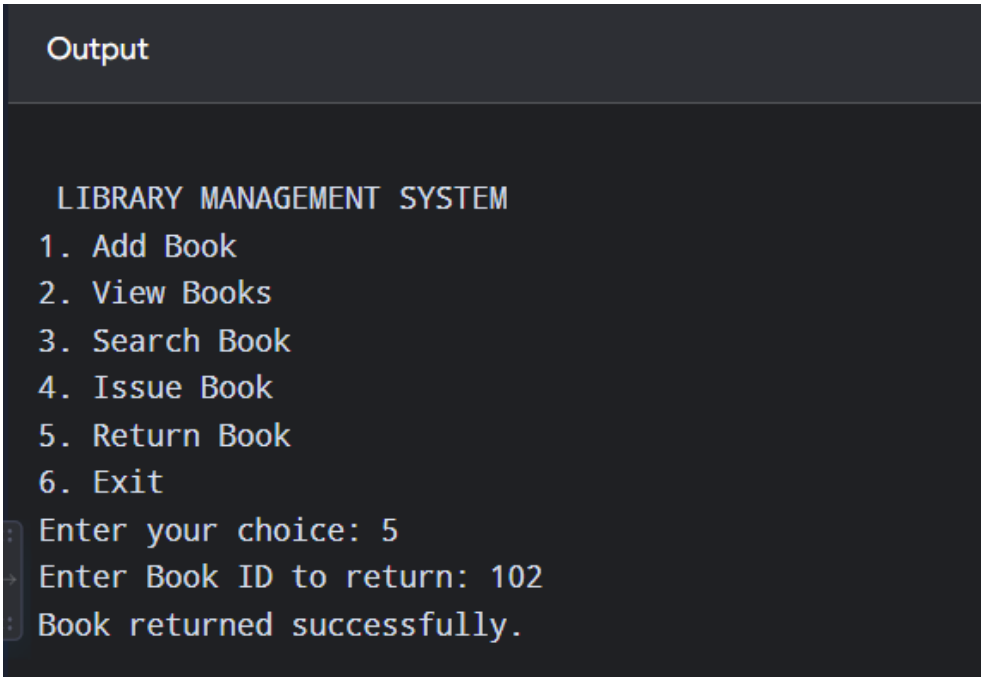
3) Search Book

```
Output

 LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 3
Enter Book ID to search: 101

Book Found
Title : Python Basics
Author: John Smith
Status: Available
```

4) Return Book



RESULT

The Library Management System was successfully developed using Python. The application successfully performs library operations such as adding, viewing, searching, issuing, and returning books. The project demonstrates the practical implementation of basic Python programming concepts in a real-world library management application.